Static Timing Analysis (STA) Across All SKY130 Timing Corners

AIM

To perform **Static Timing Analysis (STA)** using OpenSTA for the SKY130 standard-cell library and analyze the timing performance of the design at different PVT corners.

TOOLS AND FILES USED

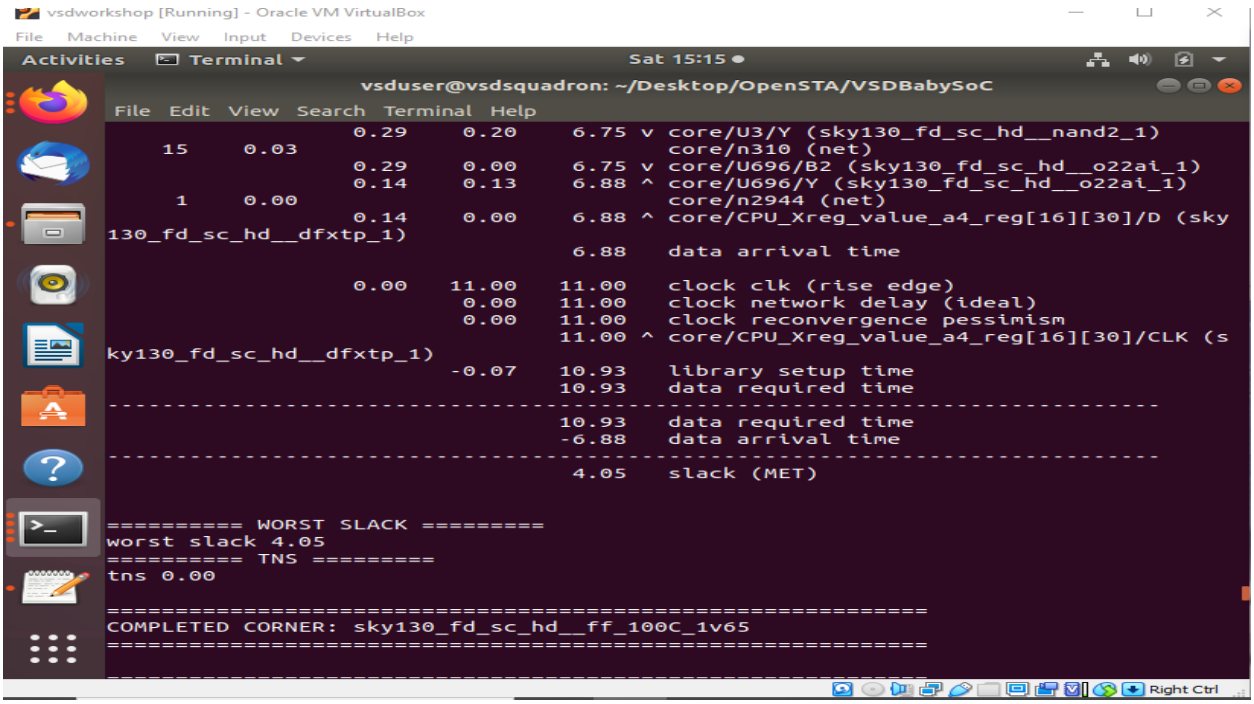
- **OpenSTA** – Static Timing Analysis
- **SKY130 standard-cell timing libraries**
- **VSDBabySoC synthesized netlist**
- **SDC timing constraints**
- **Ubuntu/Linux environment**

Input Files

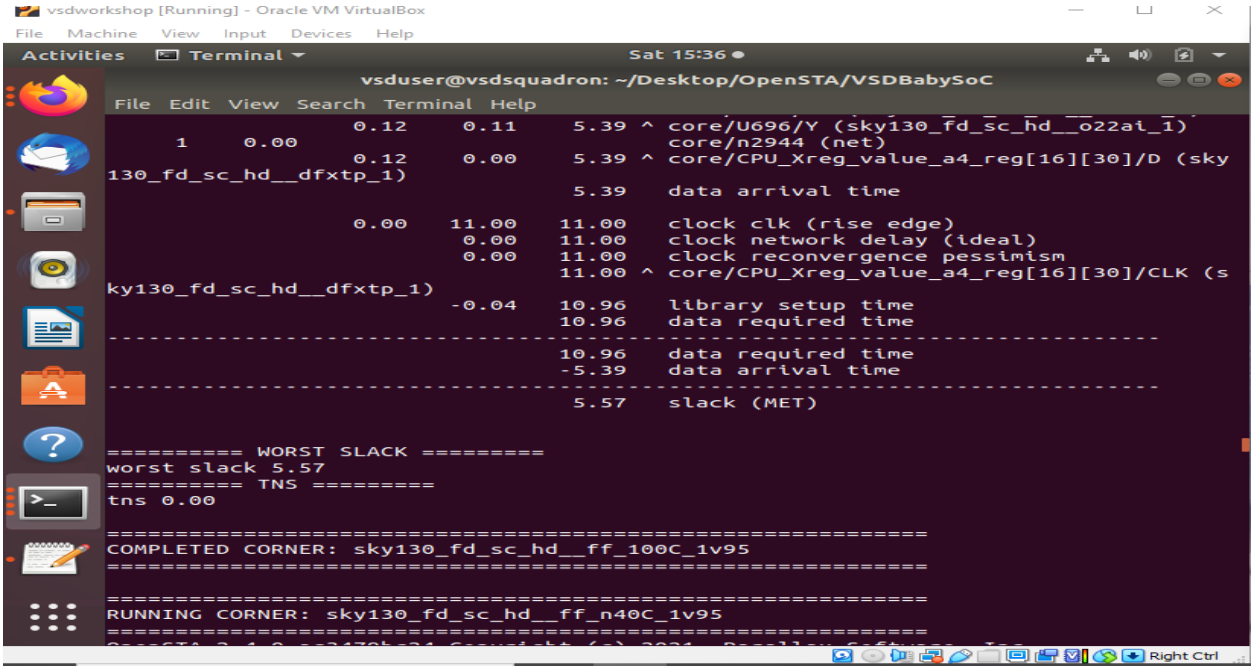
File	Purpose
`vsdbabysoc.synth.v`	Synthesized gate-level netlist
`vsdbabysoc_synthesis.sdc`	Timing constraints
`avsdpll.lib`	PLL library
`avsddac.lib`	DAC library
`*.lib`	SKY130 timing corner libraries

1. FF CORNERS

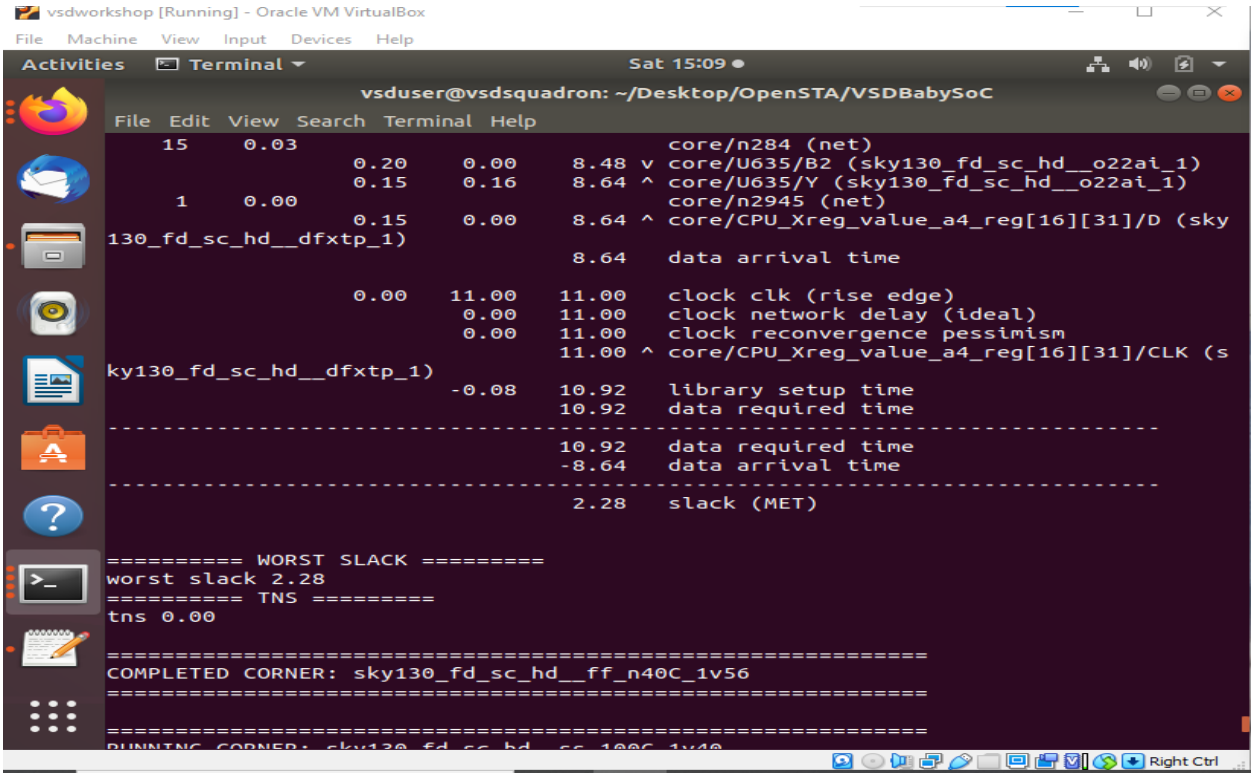
- `sky130_fd_sc_hd__ff_100C_1v65.lib`



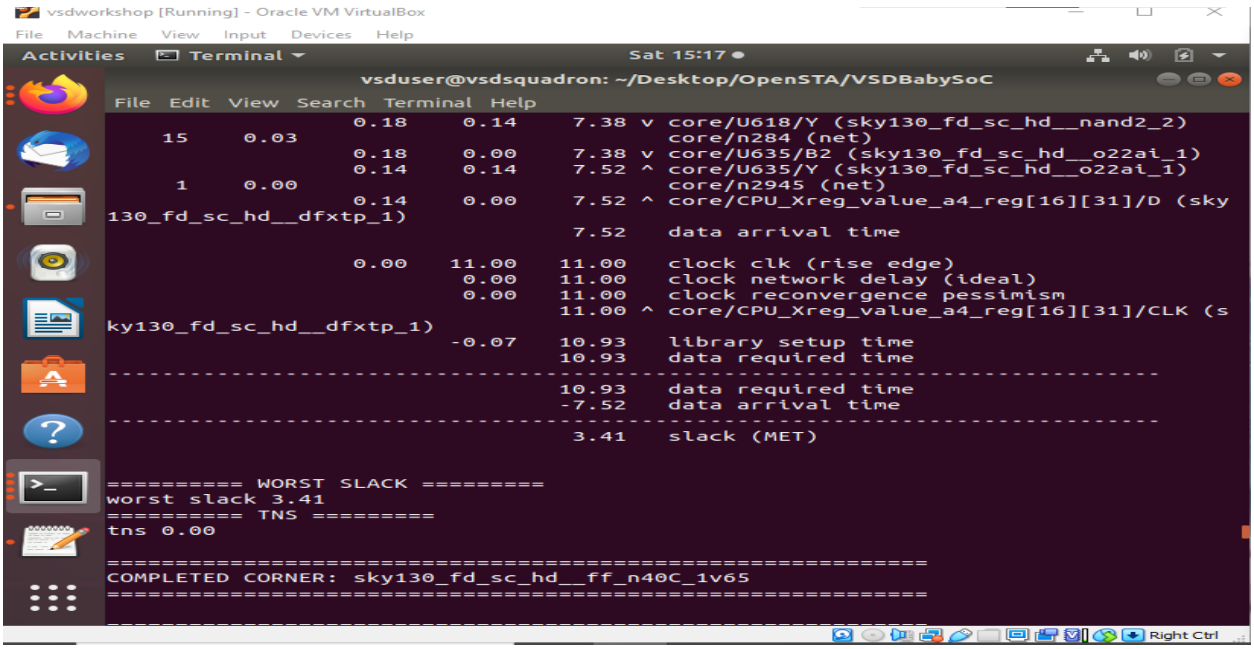
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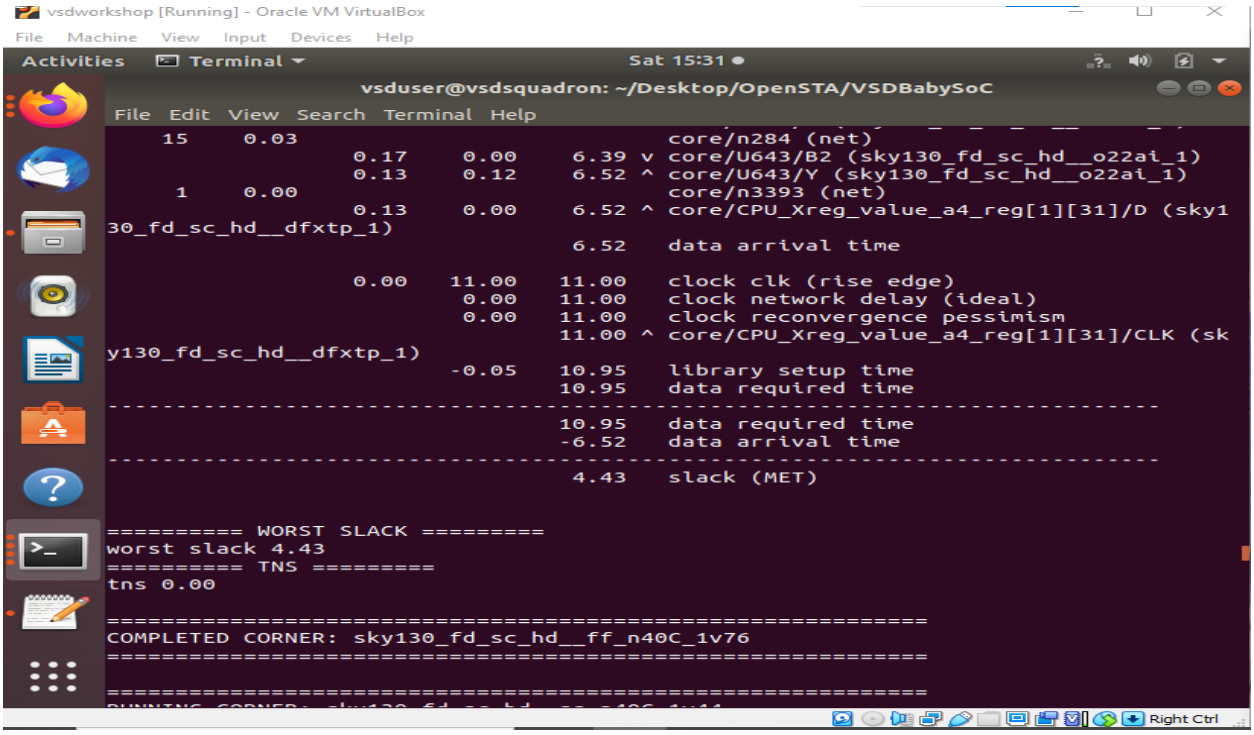
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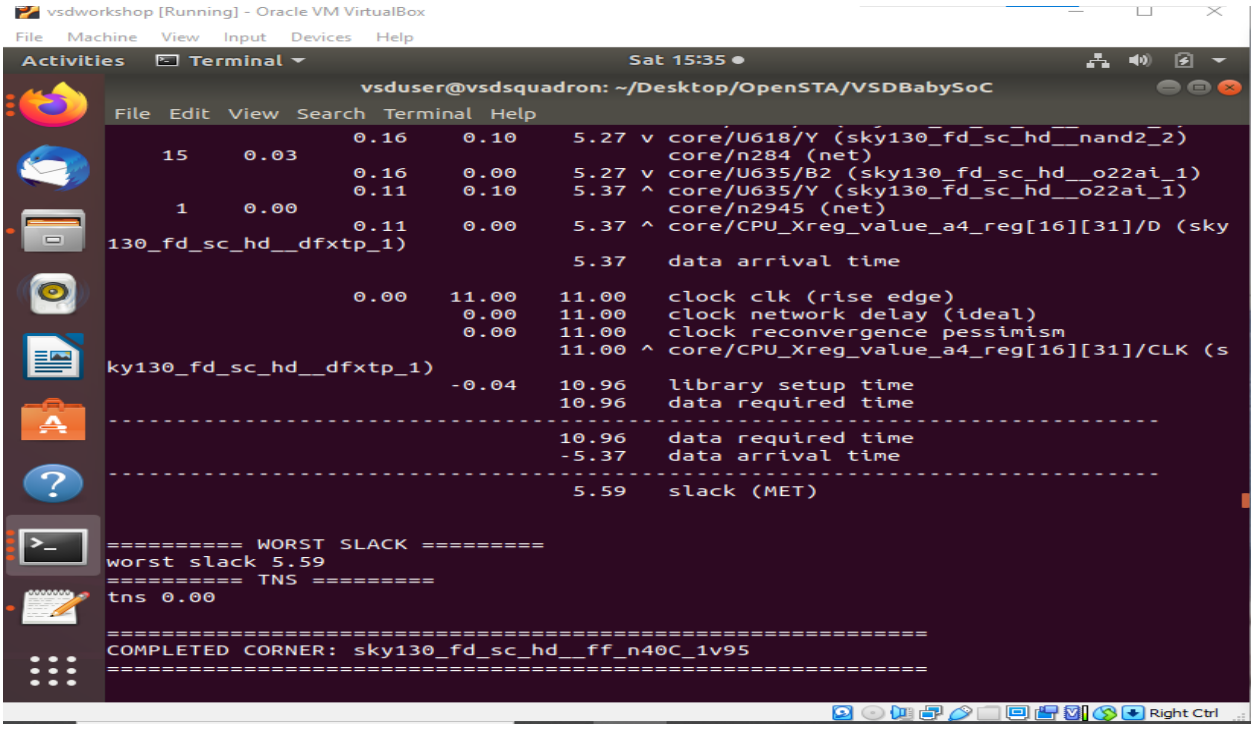
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sky130_fd_sc_hd__ff_n40C_1v76.lib

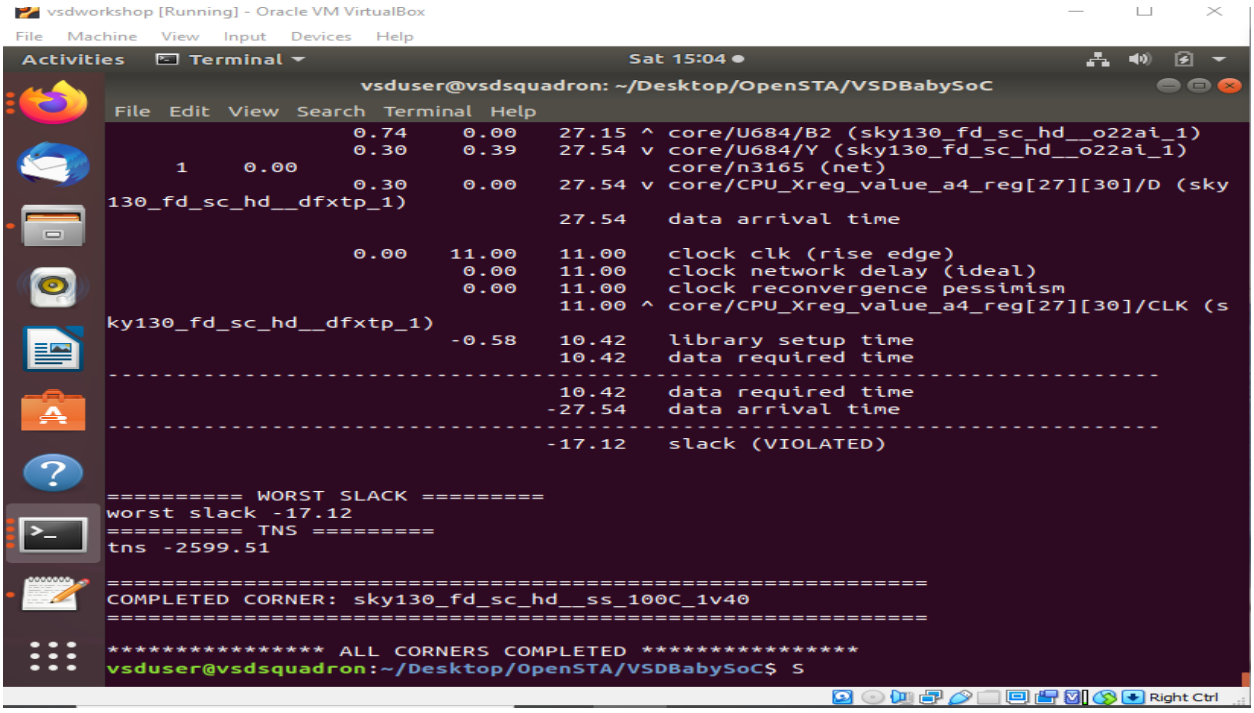


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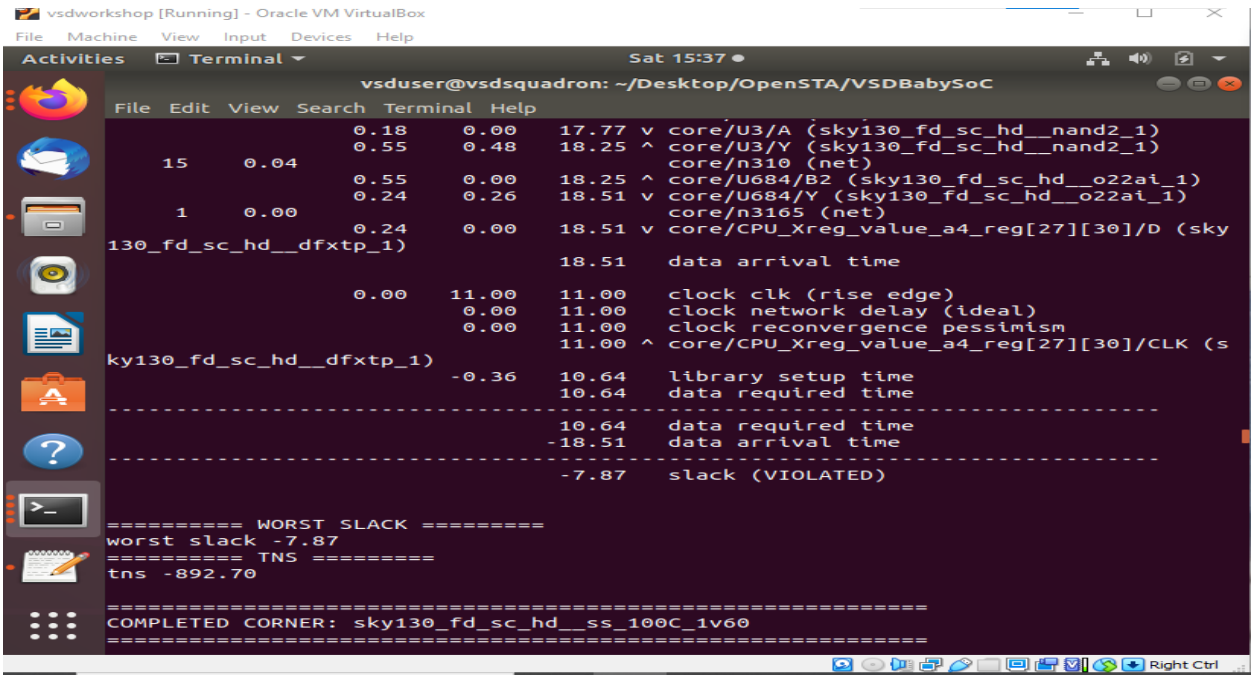


SS CORNERS

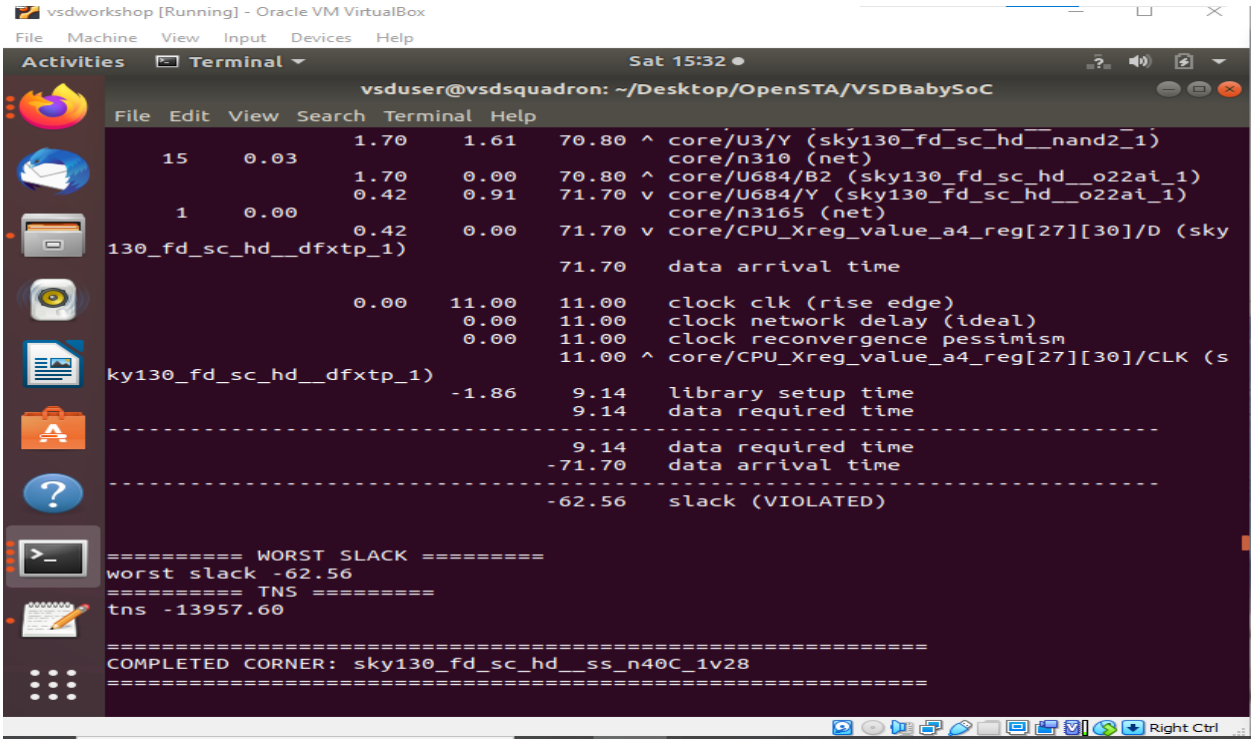
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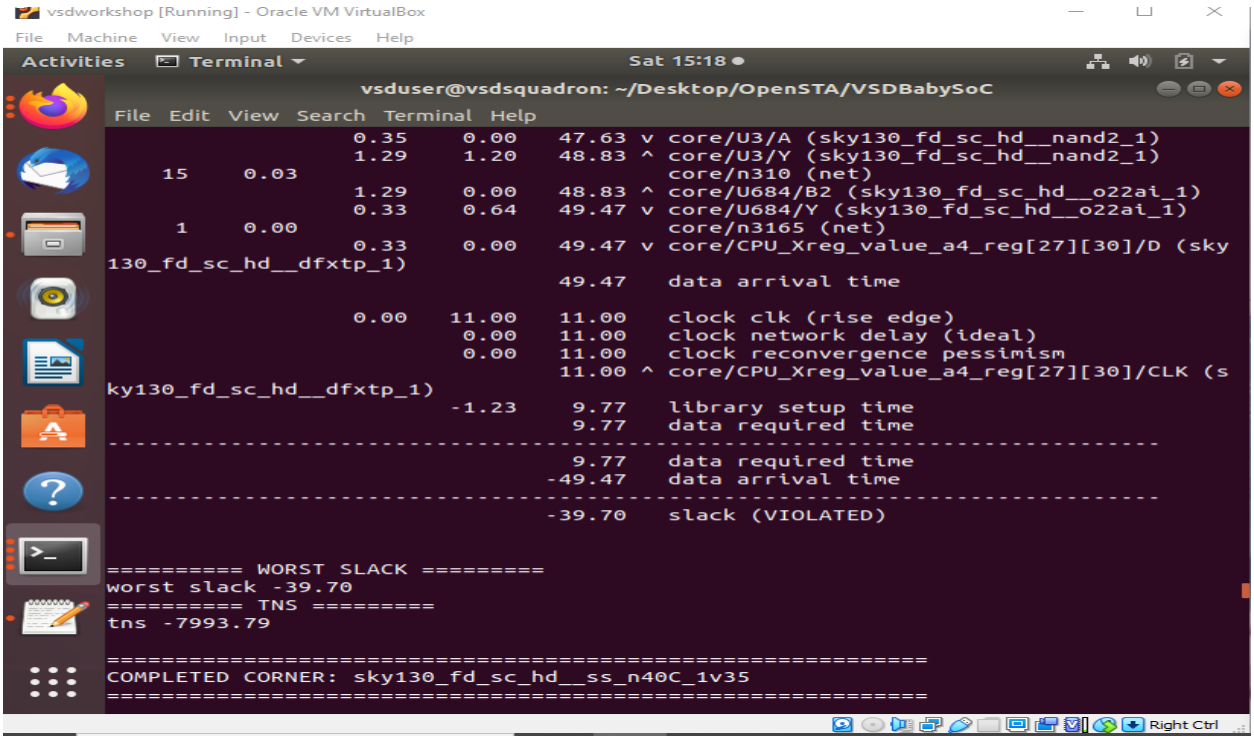
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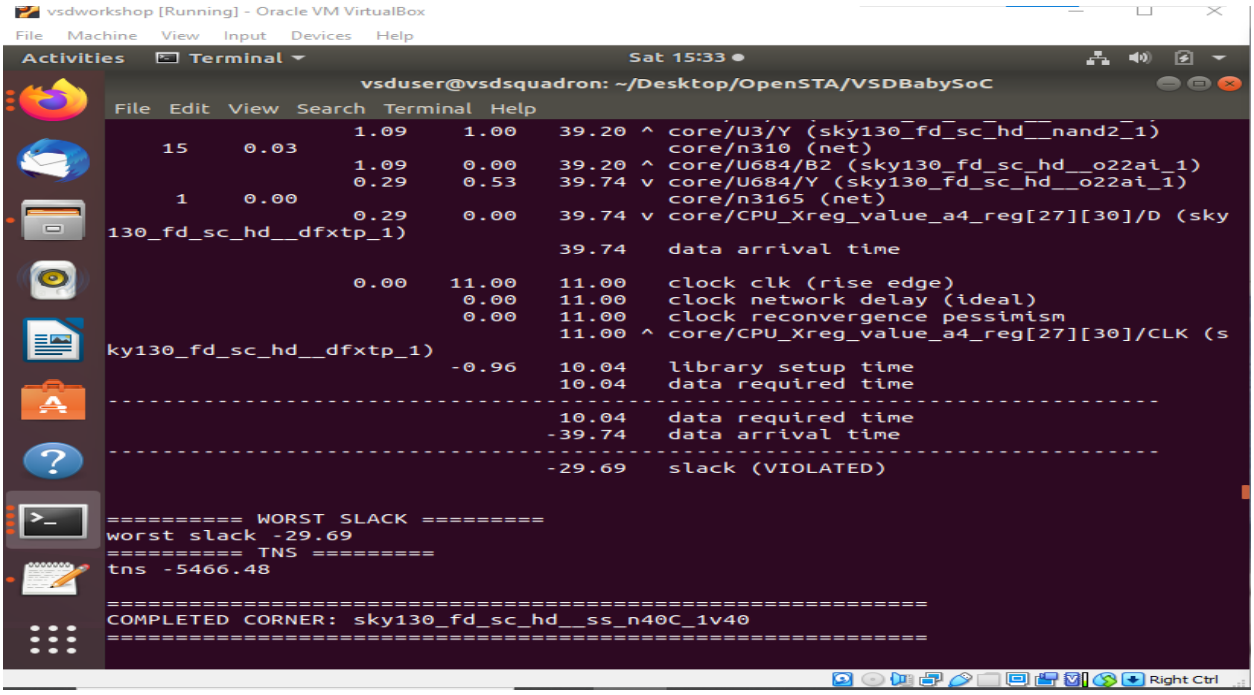
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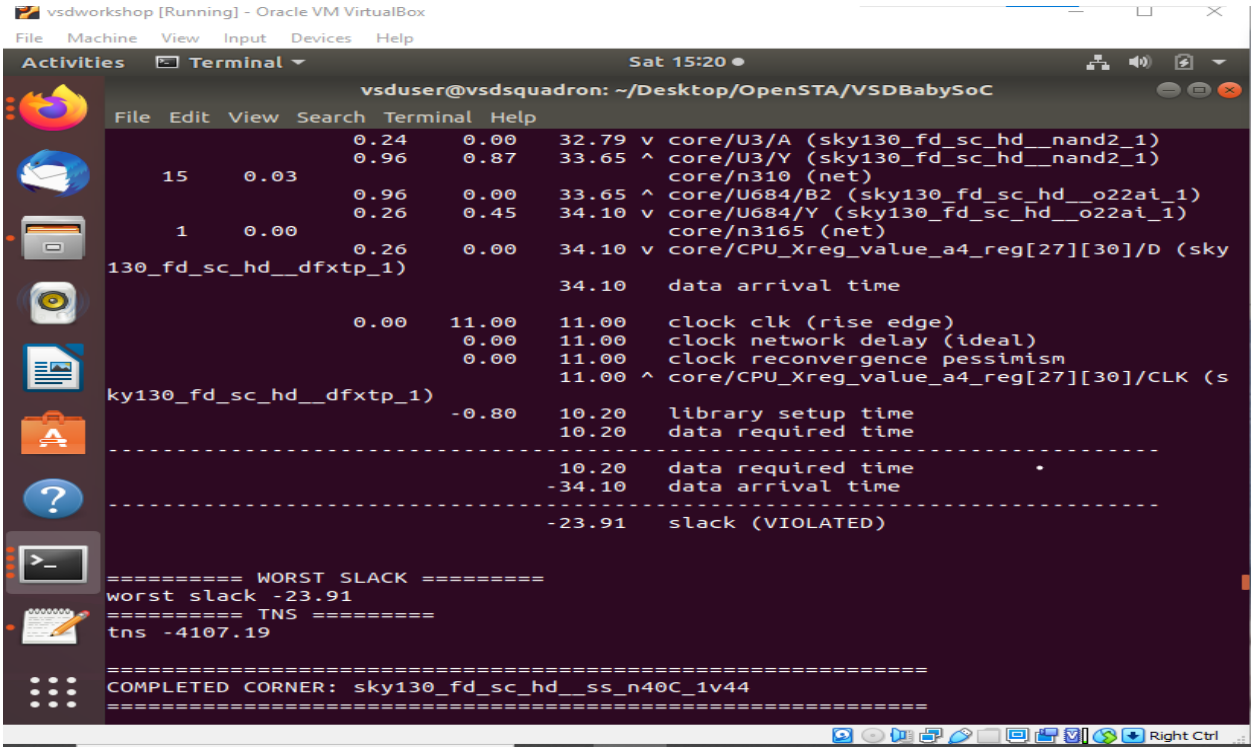
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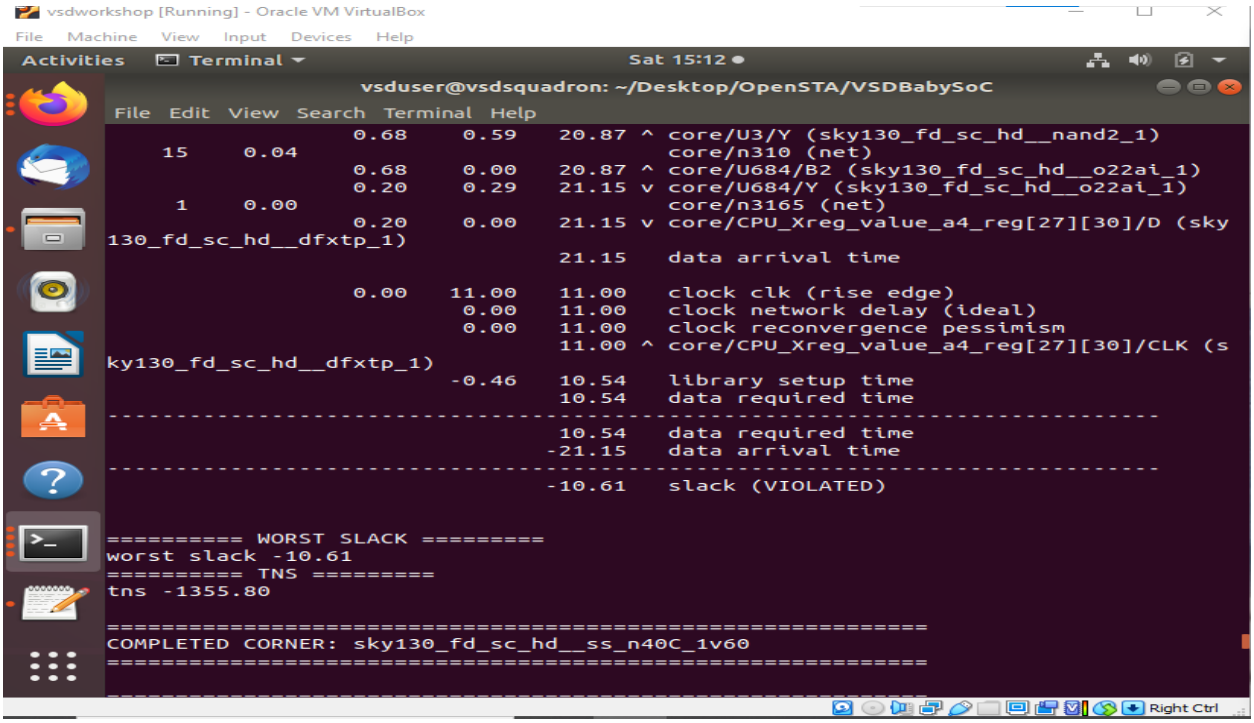
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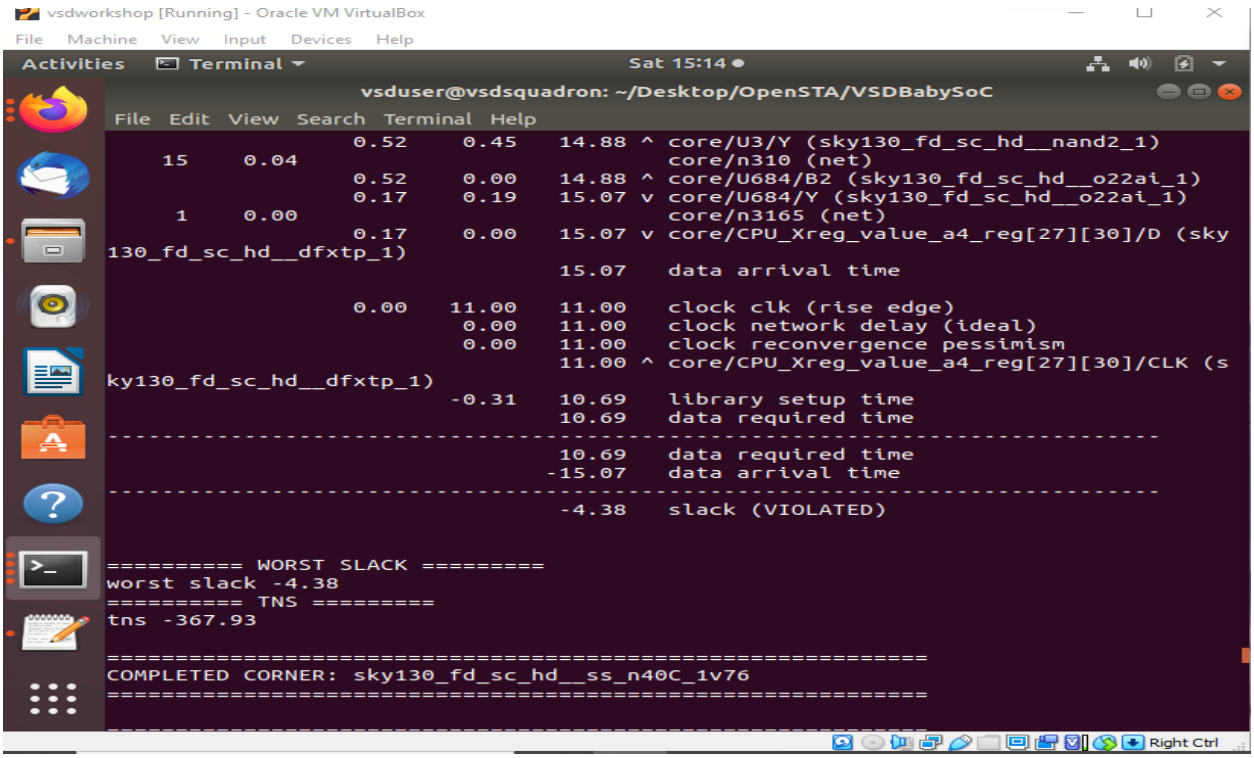
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sky130_fd_sc_hd__ss_n40C_1v60.lib

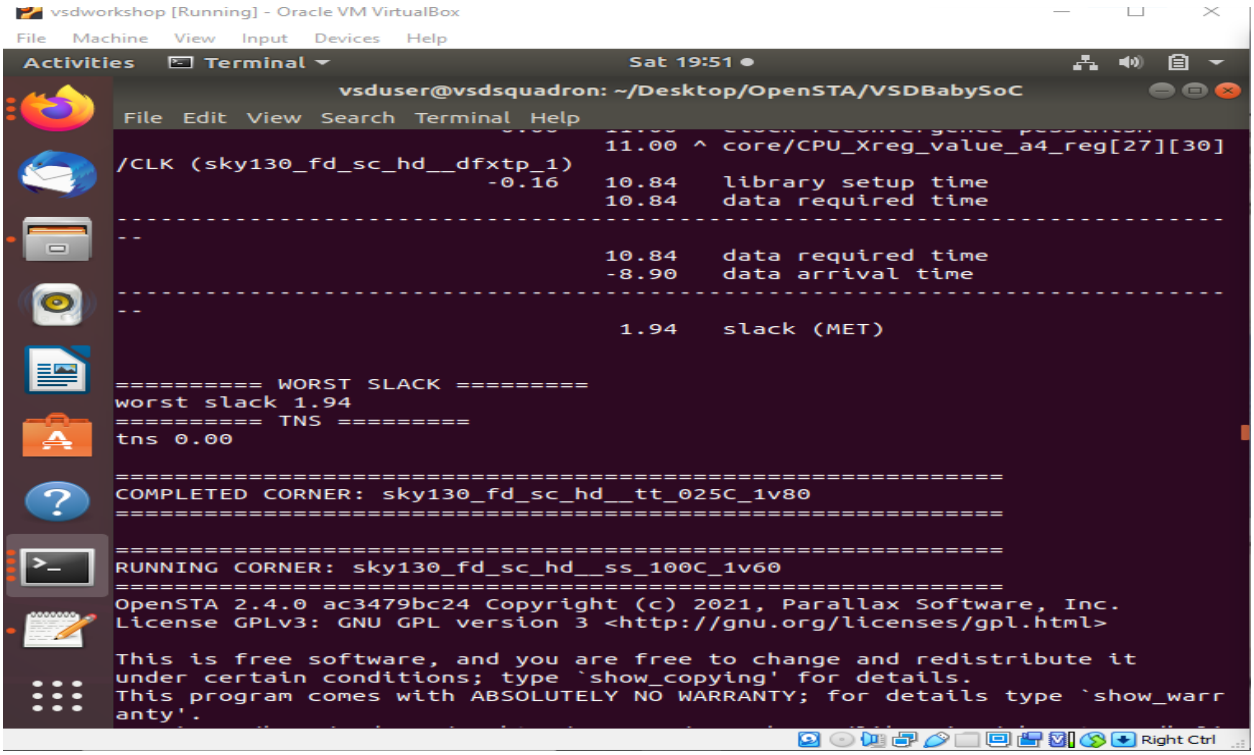


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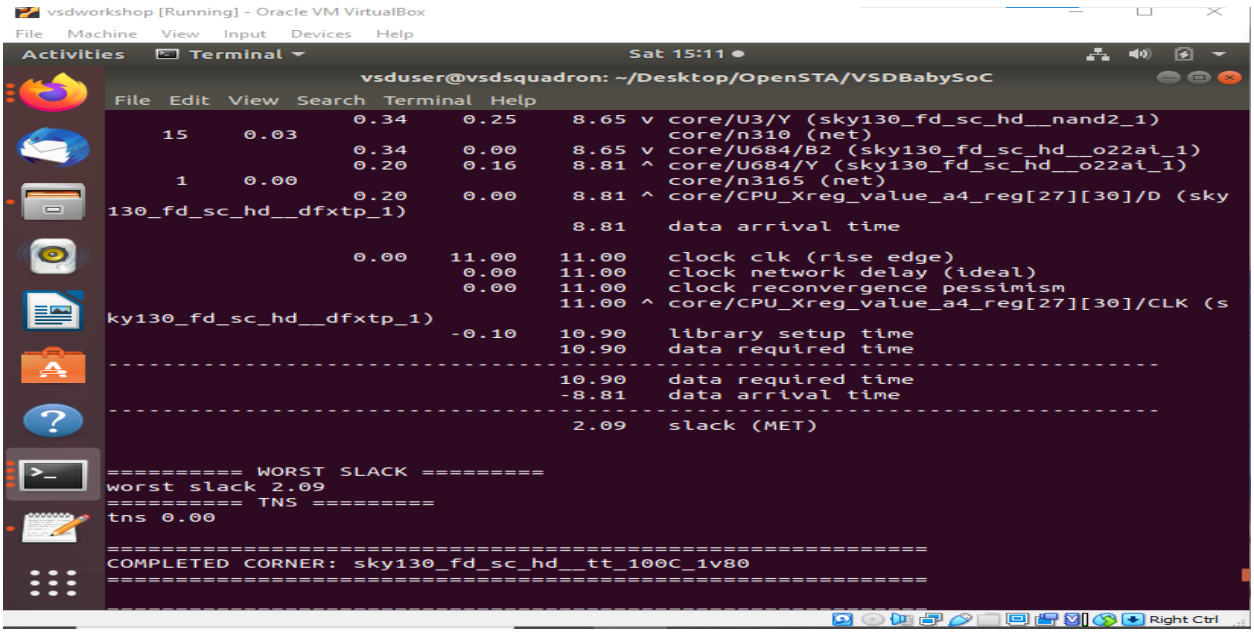


TT CORNERS

sky130_fd_sc_hd__tt_025C_1v80.lib



sky130_fd_sc_hd__tt_100C_1v80.lib



EXCEL RESULTS

STA_Sky130_TNS_WNS_with_Slack_and_labels

A1 Sky130 Liberty Corner

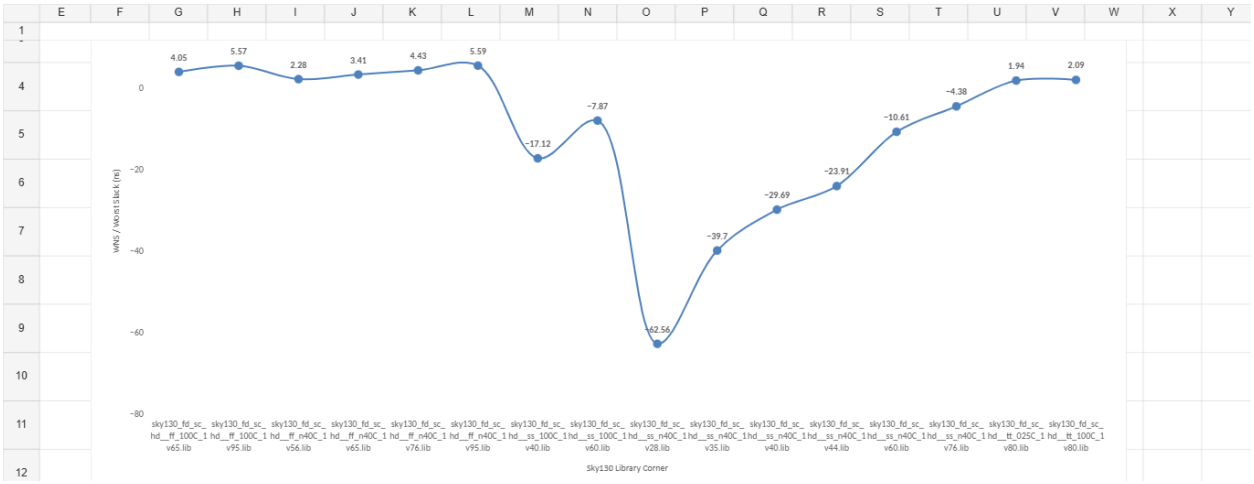
	A	B	C	D
1	Sky130 Liberty Corner	TNS (ns)	WNS (ns)	Slack (ns)
2	sky130_fd_sc_hd__ff_100C_1v65.lib	0.00	4.05	4.05
3	sky130_fd_sc_hd__ff_100C_1v95.lib	0.00	5.57	5.57
4	sky130_fd_sc_hd__ff_n40C_1v56.lib	0.00	2.28	2.28
5	sky130_fd_sc_hd__ff_n40C_1v65.lib	0.00	3.41	3.41
6	sky130_fd_sc_hd__ff_n40C_1v76.lib	0.00	4.43	4.43
7	sky130_fd_sc_hd__ff_n40C_1v95.lib	0.00	5.59	5.59
8	sky130_fd_sc_hd__ss_100C_1v40.lib	-2599.51	-17.12	-17.12
9	sky130_fd_sc_hd__ss_100C_1v60.lib	-892.70	-7.87	-7.87
10	sky130_fd_sc_hd__ss_n40C_1v28.lib	-13957.60	-62.56	-62.56
11	sky130_fd_sc_hd__ss_n40C_1v35.lib	-7993.79	-39.70	-39.70
12	sky130_fd_sc_hd__ss_n40C_1v40.lib	-5466.48	-29.69	-29.69
13	sky130_fd_sc_hd__ss_n40C_1v44.lib	-4107.19	-23.91	-23.91
14	sky130_fd_sc_hd__ss_n40C_1v60.lib	-1355.80	-10.61	-10.61
15	sky130_fd_sc_hd__ss_n40C_1v76.lib	-367.93	-4.38	-4.38
16	sky130_fd_sc_hd__tt_025C_1v80.lib	0.00	1.94	1.94
17	sky130_fd_sc_hd__tt_100C_1v80.lib	0.00	2.09	2.09
18				
19				
20				
21				
22				
23				
24				
25				

README STA Data TNS Chart WNS Chart

WNS Graph

WNS(Worst Negative Slack) **Across SKY130 PVT Corners**

X-axis = SKY130 Timing Corner
Y-axis = WNS (ns)



TNS Graph

TNS(Total Negative Slack) **Across SKY130 PVT Corners**

X-axis = SKY130 Timing Corner
Y-axis = TNS (ns)

